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Progress Report:

Deep Representation of Transcriptomics and Radiomics in Predicting Treatment Outcomes in Hepatocellular Carcinoma

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Contents

Chapter 1

Introduction

Hepatocellular carcinoma (HCC) is the leading cause of primary liver cancer and the third-leading cause of cancer-related mortality worldwide. For patients diagnosed with early-stage HCC, surgical resection offers a potentially curative treatment pathway; however, this is severely undermined by high post-surgical recurrence rates, affecting approximately 70% of patients within five years [1, 2]. Recurrence typically follows a dual pattern: early recurrence driven by occult metastasis, and late recurrence emerging *de novo* from the underlying carcinogenic cirrhotic liver field [2].
While clinical staging algorithms such as the Barcelona Clinic Liver Cancer (BCLC) system guide initial treatment decisions, they lack the granular molecular information necessary to stratify individual recurrence risk [1]. Genomic and transcriptomic profiling of both the tumour and adjacent cirrhotic tissue can independently predict these recurrence patterns [2], yet acquiring tissue for molecular assays requires invasive biopsies that carry significant bleeding and complication risks in patients with advanced chronic liver disease, limiting routine clinical implementation.
In contrast, pre-treatment radiological imaging (CT and MRI) is non-invasive and universally acquired as the standard of care for HCC staging. Recent work demonstrates that quantitative imaging biomarkers (radiomics [3]) can predict HCC clinical outcomes including survival and immunotherapy response, outperforming standard clinical parameters such as BCLC stage and ALBI grade [4]. However, traditional radiomics rely on predefined, hand-crafted features and supervised learning, which require substantial expert labelling and may fail to capture the full mesoscopic complexity of the tumour microenvironment.
This project proposes to bridge non-invasive imaging and molecular phenotyping by using transformer-based vision foundation models to learn a shared latent space between pre-treatment MRI and tumour transcriptomic profiles. We hypothesise that self-supervised vision models applied to routine MRI can learn latent representations that capture the underlying molecular phenotype of HCC, enabling prediction of post-resection recurrence risk without invasive tissue sampling.

Chapter 2

Background

2.1 Radiomics and Clinical Outcome Prediction

Radiomics refers to the high-throughput extraction of quantitative features from medical images, encompassing shape, texture, and wavelet transformations [3]. Applied to HCC, machine learning models trained on radiomic features extracted from pre-treatment CT and MRI have demonstrated the ability to predict survival, immunotherapy response, and tumour biology, outperforming conventional clinical biomarkers in several cohorts [4]. Despite their promise, traditional radiomic pipelines are inherently limited by their dependence on manually engineered feature definitions, which may not generalise across acquisition protocols or capture subtle heterogeneity in the tumour microenvironment.

2.2 Foundation Models and Vision Transformers

The field of medical AI is shifting toward Generalist Medical AI (GMAI) [5], in which foundation models leverage self-supervised learning to discover structural patterns from large, diverse datasets without the need for task-specific labels. Vision Transformers (ViTs) trained via DINO-style self-supervised learning can extract robust, general-purpose visual embeddings without manual feature engineering [6, 7]. By utilising discriminative self-supervised objectives, these models learn representations at both the global image and local patch levels, capturing pixel-level tissue architectures and subtle heterogeneities that predefined radiomic descriptors miss [6]. Because the learned features are task-agnostic and generalise across varied image distributions, they are well-suited to capture the deep radiological phenotype of HCC directly from standard MRI.

2.3 Multimodal Alignment: Contrastive Learning and DINO

Contrastive learning provides a principled framework for aligning representations across modalities. The SimCLR / NT-Xent framework [8] maximises agreement between positive pairs (here, image and gene embeddings of the same patient) while pushing apart negative pairs, learning a shared latent space without explicit supervision.
The quality of the shared space depends critically on the vision backbone. DINO [7] and DINOv2 [6] train a student ViT to match an exponential-moving-average teacher via a self-distillation objective, yielding patch-level features that spontaneously cluster into semantically coherent regions and a global `[CLS]` token that transfers broadly across tasks. These structured representations reduce the cross-modal representational gap: Girdhar et al. [9] showed that a strong DINO-anchored vision encoder can serve as a universal embedding hub, binding six heterogeneous modalities transitively without pairwise training. In medical imaging, the same principle has been applied to align pathology tile embeddings with clinical text signatures [10]. For MRI–transcriptomics alignment in HCC, a DINO-pretrained backbone therefore provides a stronger prior than a generic ImageNet encoder, mitigating patient-identity memorisation and improving the fidelity of learned transcriptomic surrogates.

2.4 Virtual Biopsy

The feasibility of inferring molecular profiles from imaging data has been demonstrated in digital pathology, where deep learning models predict bulk RNA-seq expression profiles directly from histological whole-slide images [11]. Extending this paradigm to macroscopic radiology, recent work has shown that deep multimodal architectures can learn a shared latent space between CT scans and tissue metabolomics in non-small-cell lung cancer, enabling non-invasive classification and prognostication [12]. This project aims to translate and advance this approach for HCC by aligning pre-treatment MRI embeddings with transcriptomic profiles.

2.5 Project Objectives

The main objectives of this project are:

1. **Foundation model image embeddings:** Use transformer-based vision foundation models (ViTs with contrastive / self-supervised learning) to derive high-dimensional embeddings from pre-treatment MRI scans, and benchmark them against a radiomics feature baseline.

2. **Virtual transcriptomic profiling:** Use the shared latent space to infer transcriptomic representations from radiological data alone, enabling non-invasive approximation of tumour molecular profiles.

3. **Outcome prediction:** Predict post-resection treatment outcomes and tumour recurrence directly from imaging-derived and inferred transcriptomic embeddings, comparing foundation model representations against radiomics-only baselines.

Chapter 3

Experiments

3.1 Baselines for RFS Prediction

Methods

Cohort

Baselines for recurrence-free survival (RFS) prediction were evaluated on the resection cohort. Patients without a recorded event and with follow-up shorter than the prediction horizon were excluded, giving 1-year RFS $n = 56$ (34% recurrence) and 2-year RFS $n = 54$ (48% recurrence). Clinical and pathological characteristics of the cohort are given in Table 3.1.

Evaluation

Models were evaluated using 3-fold stratified cross-validation (CV). Classifiers were L_1-regularised logistic regression (LR; $C \in \{0.001, 0.01, 0.1, 1\}$) and random forest (RF; depth $\in \{2, 4\}$, min leaf $\in \{5, 10, 15\}$). AUROC of the best hyperparameter configuration per fold was reported, with mean and standard deviation across folds.

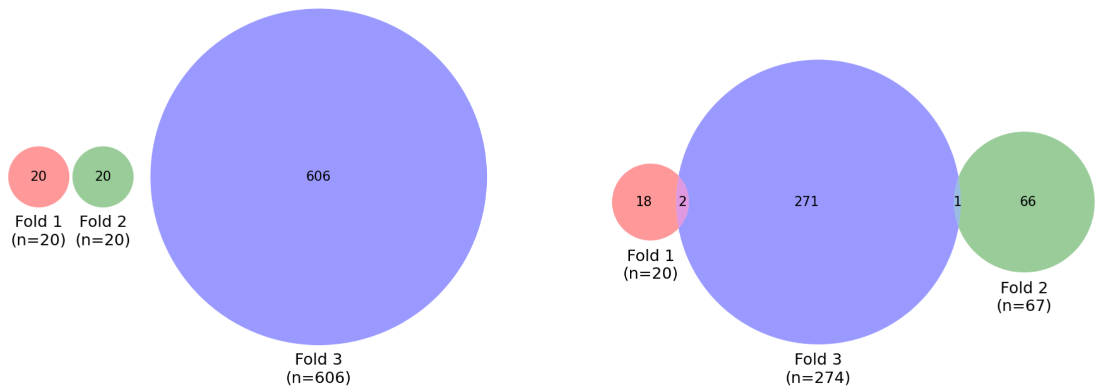
Modalities and Features

Two modalities were assessed. *RNA-seq*: DESeq2 [13] differential-expression analysis (Benjamini–Hochberg adjusted $p < 0.05$; fallback to the top 20 genes by adjusted p-value when fewer than 20 passed the threshold) applied to two input sets: (i) the full transcriptome (50,986 genes) and (ii) a predefined HCC gene set (2,146 genes curated from published HCC literature). In-CV DESeq2 selection (run independently inside each training fold) is the statistically valid approach but proved highly unstable at this sample size: with approximately 36 training patients per fold, zero genes overlapped between folds at either time horizon (Figure 3.1). To obtain a more stable feature set for comparison and for informing the multimodal experiments, we also ran DESeq2 once on the full task-filtered dataset before cross-validation; the resulting gene lists are given in Appendix A and informed the gene-set design in Section 3.2.

Table 3.1: Clinical and pathological characteristics of the resection cohort ($n = 60$). TTR = time-to-recurrence; RFS = recurrence-free survival; OS = overall survival.

		All patients ($n = 60$)
Demographics	Age, mean ($\pm$SD), years	62.7 ± 12.1
	Male	53 (88.3%)
	Female	7 (11.7%)
ECOG PS	0	60 (100%)
HCC Etiology	Hepatitis B	13 (21.7%)
	Hepatitis C	13 (21.7%)
	Alcohol	15 (25.0%)
	NASH	10 (16.7%)
Tumour	No. of lesions, mean ($\pm$SD)	1.7 ± 1.2
	Max diameter, mm, mean ($\pm$SD)	43.0 ± 26.5
BCLC Stage	A	8 (13.3%)
	B	41 (68.3%)
	C	11 (18.3%)
Child-Pugh	A	55 (91.7%)
	B	2 (3.3%)
Child-Pugh Points	5	41 (68.3%)
	6	14 (23.3%)
	7	2 (3.3%)
Vascular Invasion	None	20 (33.3%)
	Micro	35 (58.3%)
	Micro/macro	2 (3.3%)
Resection Margin	R0	37 (61.7%)
	RFA	10 (16.7%)
	Macro/micro	6 (10.0%)
	Macro	2 (3.3%)
Tumour Grade	Well	15 (25.0%)
	Moderate	39 (65.0%)
	Poor	6 (10.0%)
Survival	TTR: events (n)	34
	median time-to-event, months	11.6
	median follow-up (censored), months	36.9
	RFS: events (n)	41
	median time-to-event, months	14.3
	median follow-up (censored), months	52.7
	OS: events (n)	21
	median time-to-event, months	52.7
	median follow-up (censored), months	44.3

Arterial-phase radiomics: 149 features extracted from the arterial-phase MRI, with in-CV and before-CV feature selection (`SelectKBest` with F-score, $k = 100$).



(a) 1-year RFS **(b)** 2-year RFS

Figure 3.1: Gene overlap across the three CV folds for in-CV DESeq2 selection. Zero genes are shared across all three folds at either time horizon, confirming the instability of in-CV selection at this sample size.

Results

Unimodal Performance

Table 3.2 summarises 3-fold CV AUROC for both modalities across 1-year and 2-year RFS under in-CV and before-CV feature selection.

Table 3.2: 3-fold CV AUROC (mean $\pm$ std) for arterial radiomics and RNA-seq baselines. Best classifier (LR or RF) shown per row.

	1-year RFS		2-year RFS	
Modality	In-CV	Before-CV	In-CV	Before-CV
Arterial radiomics	0.718 ± 0.070	0.821 ± 0.068	0.569 ± 0.133	0.781 ± 0.069
RNA-seq, full transcriptome	0.492 ± 0.089	0.779 ± 0.083	0.590 ± 0.191	0.867 ± 0.097
RNA-seq, predefined HCC genes	0.596 ± 0.133	0.596 ± 0.133	0.624 ± 0.038	0.591 ± 0.071

On in-CV evaluation, arterial radiomics clearly outperformed RNA-seq at 1-year, confirming that imaging features capture short-term recurrence risk more reliably than transcriptomics at this cohort size. At 2-year, the gap narrowed for in-CV feature selection, and RNA-seq outperformed radiomics under before-CV selection, with a mean AUROC of 0.867 versus 0.781, suggesting that RNA-seq carries stronger prognostic signal for longer-term recurrence.

For the predefined HCC gene set, no gene survived BH correction at $p < 0.05$ or $p < 0.1$ in any fold or configuration: every run collapsed to the 20-gene fallback (the top 20 by adjusted p-value, which were effectively the first columns in matrix order when all $p_{\text{adj}} \to 1$). Though the predefined HCC gene set had significantly lower AUC compared to the full transcriptome when using a before-CV selector, it still outperformed with an in-CV selector. Thus, we still included the predefined HCC gene set in the multimodal experiments.

Late-Fusion Ensemble

To examine the potential complementarity between the two modalities, a late-fusion ensemble was constructed by averaging predicted probabilities from the best before-CV models per modality (radiomics RF with max depth 2; RNA-seq LR with $C = 1$).

Table 3.3: 2-year RFS AUROC per fold and mean $\pm$ std for the late-fusion ensemble versus individual modalities (before-CV selection).

Fold	Ensemble	Radiomics (RF)	RNA-seq (LR)
1	0.812	0.688	0.750
2	0.963	0.802	0.988
3	0.901	0.852	0.864
Mean $\pm$ std	$\mathbf{0.892 \pm 0.062}$	0.781 ± 0.069	0.867 ± 0.097

The ensemble outperformed both individual modalities in every fold, reaching a mean AUROC of 0.892 versus 0.781 for radiomics alone ($+0.111$) and 0.867 for RNA-seq alone ($+0.025$). The consistent per-fold improvement demonstrates that transcriptomic and radiomic features carry complementary prognostic information for 2-year recurrence.
To better control for demographic factors, age and sex were added as demographic confounders by concatenating them to the selected features after feature selection and before model fitting. Table 3.4 shows the per-fold results with this adjustment.

Table 3.4: 2-year RFS AUROC per fold and mean $\pm$ std for the late-fusion ensemble versus individual modalities, with Age and Sex as additional confounders (before-CV selection).

Fold	Ensemble	Radiomics (RF)	RNA-seq (LR)
1	0.800	0.588	0.775
2	0.963	0.840	0.988
3	0.914	0.901	0.864
Mean $\pm$ std	$\mathbf{0.892 \pm 0.068}$	0.776 ± 0.136	0.876 ± 0.087

The mean ensemble AUROC is unchanged at 0.892. Adding Age and Sex marginally shifts the individual modalities (radiomics RF: $0.781 \to 0.776$, -0.005; RNA-seq LR: $0.867 \to 0.876$, $+0.009$), with the two changes cancelling at the ensemble level.

Summary

Taken together, 1-year recurrence is dominated by imaging features while 2-year prediction benefits from integrating RNA-seq data. These findings motivate the focus on 2-year RFS and the multimodal experiments in Sections 3.2 and 3.3.

3.2 Multimodal Contrastive Learning

Methods

Cohort

Contrastive models were pre-trained on the resection cohort with 2-year RFS labels ($n = 54$).

Image Preprocessing

Raw arterial-phase MRI volumes were resampled to $1\times1\times3\,\text{mm}$ isotropic before training, with no further intensity pre-processing beyond per-slice normalisation.

Model Architecture

The image encoder used a ViT-B/32 backbone (ImageNet1K pretrained) followed by a 2-layer MLP projecting to 128 dimensions; the gene encoder used a 3-layer MLP projecting to 128 dimensions.

Training

Training used NT-Xent loss ($\tau = 0.07$) with AdamW (lr=10^{-4}). 10% of the data was used for validation, and the model with the best validation loss was selected for downstream evaluation. For a batch of N paired samples, let $\mathbf{u}_i$ and $\mathbf{v}_i$ denote the ℓ_2-normalised projected image and gene embeddings of patient i. The NT-Xent loss is

$$\mathcal{L}_{\text{NT-Xent}} = -\frac{1}{2N}\sum_{i=1}^{N}\left[\log\frac{e^{\mathrm{sim}(\mathbf{u}_i,\mathbf{v}_i)/\tau}}{\sum_{j=1}^{N} e^{\mathrm{sim}(\mathbf{u}_i,\mathbf{v}_j)/\tau}} + \log\frac{e^{\mathrm{sim}(\mathbf{v}_i,\mathbf{u}_i)/\tau}}{\sum_{j=1}^{N} e^{\mathrm{sim}(\mathbf{v}_i,\mathbf{u}_j)/\tau}}\right], \tag{3.1}$$

where $\mathrm{sim}(\mathbf{a},\mathbf{b}) = \mathbf{a}^\top\mathbf{b}\,/\,(\|\mathbf{a}\|\|\mathbf{b}\|)$.
Two regularisation settings were evaluated: $\lambda = 0.1$ and $\lambda = 0$ (pure contrastive). The regularisation term adds a per-modality binary cross-entropy loss with a linear classification head h,

$$\mathcal{L}_{\text{reg}} = \mathcal{L}_{\text{BCE}}\big(h(\mathbf{u}),\, y\big) + \mathcal{L}_{\text{BCE}}\big(h(\mathbf{v}),\, y\big), \tag{3.2}$$

where $y \in \{0,1\}$ is the 2-year RFS label, giving the total training objective

$$\mathcal{L} = \mathcal{L}_{\text{NT-Xent}} + \lambda\,\mathcal{L}_{\text{reg}}. \tag{3.3}$$

Configurations

Configurations varied across four axes:

Slices per patient and backbone frozen or not 10 slices per patient with ViT-B/32 backbone unfrozen versus all slices per patient withViT-B/32 backbone frozen

Image crop Full raw MRI versus bounding-box crop padded by 10 voxels around the tumour segmentation mask.

Training split In a *slice-level split*, individual sagittal slices were assigned randomly to train or validation; in a *patient-level split*, all slices from a given patient were assigned together, stratified by 2-year RFS outcome. With $n = 54$ patients and a 10% validation fraction, the patient-level strategy holds out 6 complete patients per split, whereas the slice-level strategy allows anatomy from up to 38 of 54 patients to appear in both train and validation sets simultaneously.

Regularisation $\lambda = 0.1$ versus $\lambda = 0$.

Gene ablation set Three sets were compared: (i) a 40-gene union (20 DESeq2-selected genes for 2-year RFS and 20 from the predefined HCC set), (ii) 20 genes selected by DESeq2 before CV, and (iii) 20 genes from the predefined HCC set that passed in-CV selection.

Downstream Evaluation

Downstream evaluation used 3-fold stratified CV on 2-year RFS. radiomics features and embeddings from the contrastive model were extracted for all patients, and the same CV splits were applied to both modalities. Feature selection was performed using $\texttt{SelectKBest}(f_{\text{classif}}, k = 100)$ within each fold, followed by LR with L1 regularisation or RF classifiers.

Results

Table 3.5: 3-fold in-CV AUC (mean ± std) on 2-year RFS, resection cohort ($n = 54$), all-slice inference. Best of LR or RF shown per row.

Configuration	Slice split		Patient split	
	$\lambda = 0.1$	$\lambda = 0$	$\lambda = 0.1$	$\lambda = 0$
Radiomics (RF)	0.569 ± 0.133			
Raw MRI, unfrozen, 10 slices	0.672 ± 0.090	0.717 ± 0.087	0.690 ± 0.036	0.739 ± 0.100
Bbox, unfrozen, 10 slices	0.965 ± 0.026	0.657 ± 0.100	0.661 ± 0.117	0.568 ± 0.101
Raw MRI, Frozen, all slices	1.000 ± 0.000	0.739 ± 0.129	0.640 ± 0.038	0.648 ± 0.100

Table 3.6: Gene set ablation: 3-fold in-CV AUC (mean ± std) on 2-year RFS ($n = 54$), $\lambda = 0.1$, raw MRI, unfrozen backbone, all-slice inference. Best of LR or RF shown per row.

Gene set	Slice split	Patient split
40-gene union	0.672 ± 0.090	0.690 ± 0.036
DESeq2 before-CV (20 genes)	0.746 ± 0.117	0.789 ± 0.032
Predefined HCC (20 genes)	0.522 ± 0.087	0.457 ± 0.040

Leakage in high-capacity configurations

Under slice-level splitting, the frozen backbone and bounding-box configurations achieve near-perfect CV AUC (1.000 and 0.965 respectively for $\lambda = 0.1$). Correcting to a patient-level split collapses these to 0.640 and 0.661, exposing both as leakage artefacts driven by patient-identity memorisation. Figure 3.2 reveals the mechanism: under a slice-level split (Figure 3.2a, red), the frozen backbone achieves near-zero validation loss at epoch 1, consistent with a high-capacity encoder that has no need to learn outcome-relevant features when slices from the same patient appear in both sets. Under the corrected patient-level split (Figure 3.2a, orange), the same architecture shows rapidly diverging validation loss, confirming that the earlier result was entirely leakage-driven.

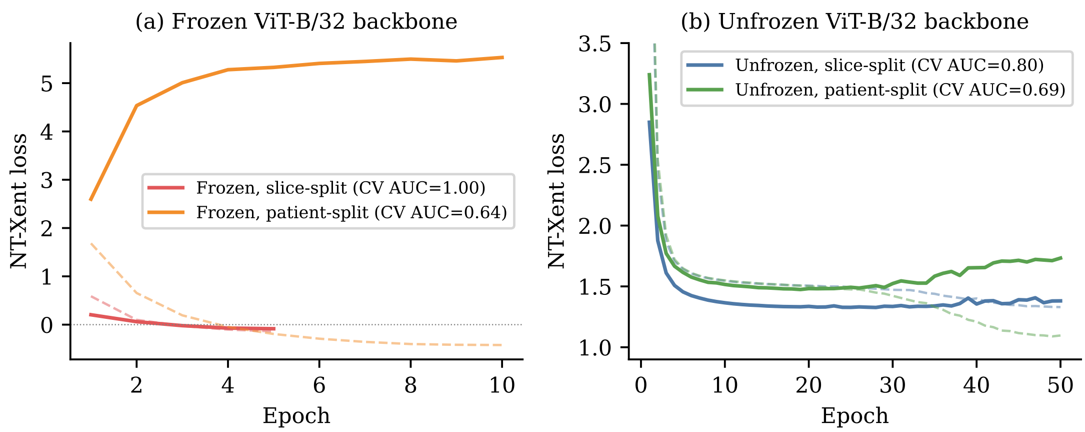


Figure 3.2: Training loss curves comparing slice-level and patient-level contrastive splits. **(a)** Frozen ViT-B/32 backbone (5–10 epochs, all slices): the slice-split model (red) achieves near-zero validation loss at epoch 1, while the patient-split model (orange) shows diverging validation loss, exposing severe memorisation under the slice split. **(b)** Unfrozen ViT-B/32 backbone (50 epochs, 10 slices): both split strategies converge similarly, with only modest divergence at later epochs under patient split. CV AUC values in the legend are for embeddings, best of LR/RF (3-fold CV).

Stability of unfrozen 10-slice models

By contrast, unfrozen 10-slice models show comparable train and validation loss trajectories under both split strategies (Figure 3.2b), and their AUC is unaffected by the correction ($\Delta \approx +0.02$; Table **??**). This stability confirms that the unfrozen models with limited number of slices learn outcome-relevant image features rather than patient anatomy, and that the patient-level evaluation is not simply a harder task — it genuinely discriminates memorisation from generalisation.

Effect of regularisation

Under honest patient-level evaluation, pure contrastive training ($\lambda = 0$) marginally outperforms the regularised variant ($\lambda = 0.1$) for raw MRI (0.739 vs. 0.690). The outcome-label regularisation term does not improve generalisation at this sample size, possibly because the linear classification head couples the embedding to RFS labels during pre-training, reducing the diversity of learned representations.

All honest evaluations beat radiomics

All patient-split configurations exceed the radiomics reference (0.569 ± 0.133), including the frozen backbone (0.640–0.648) which performed worst under honest evaluation. This indicates that contrastive pre-training on MRI yields representations with genuine prognostic value regardless of configuration choice.

Gene set ablation

Table 3.6 shows results for the gene set ablation ($\lambda = 0.1$, raw MRI, unfrozen). The DESeq2 before-CV gene set outperformed both alternatives under both split strategies. Since the before-CV selection has already contained the information of labels, we can only conclude that the contrastive model is able to leverage the prognostic signal in the DESeq2-selected genes to learn more outcome-relevant image features; Whether this can be generalised to a setting without label leakage remains to be checked in the evaluation on external cohorts in Section 3.3. The predefined HCC gene set performed poorest (0.457 patient-split), consistent with its lack of differential expression on this cohort.

3.3 External Ablation Set Evaluation

Methods

Sixteen contrastive model trained in Section 3.2 were evaluated on two independent external cohorts: the Soramic ablation set ($n = 59$, 68% positive) and the Lausanne cohort ($n = 66$, 74% positive).
The downstream head (`SelectKBest`$(k = 100)$ + LR or RF) was trained on resection embeddings and evaluated on test cohort embeddings without retraining. All MRI volumes were resampled to $1\times1\times3$ mm isotropic before embedding extraction. Radiomic baselines (149 arterial features, LR or RF) were pre-trained on the resection cohort.

Results

Table 3.7: All 16 contrastive models ranked by in-CV AUC on the resection cohort ($n = 54$), with external AUROC on Soramic ($n = 59$) and Lausanne ($n = 66$) for 2-year RFS. Parentheses show Δ versus best radiomic baseline (Soramic RF = 0.590; Lausanne LR = 0.531). *Slice-level split; all others use patient-level split.

Configuration	CV AUC	Soramic (Δ)	Lausanne (Δ)
Raw, λ=0.1, frozen, all sl.*	1.000 ± 0.000	0.718 (+0.128)	0.453 (−0.078)
Bbox, λ=0.1, unfrozen, 10 sl.*	0.965 ± 0.026	0.669 (+0.079)	0.544 (+0.013)
Raw, λ=0.1, 2y_{bCV} genes	0.789 ± 0.032	**0.732** (+0.142)	0.563 (+0.032)
Bbox, λ=0.1, frozen, all sl.*	0.766 ± 0.074	0.577 (−0.013)	0.614 (+0.083)
Raw, λ=0.1, 2y_{bCV} genes*	0.746 ± 0.117	0.516 (−0.074)	0.655 (+0.124)
Raw, λ=0.0, frozen, all sl.*	0.739 ± 0.129	0.684 (+0.094)	0.556 (+0.025)
Raw, λ=0.0, unfrozen, 10 sl.	0.739 ± 0.100	0.494 (−0.096)	0.618 (+0.087)
Raw, λ=0.0, unfrozen, 10 sl.*	0.717 ± 0.087	0.606 (+0.016)	0.600 (+0.069)
Raw, λ=0.1, unfrozen, 10 sl.	0.690 ± 0.036	0.522 (−0.068)	**0.771** (+0.240)
Raw, λ=0.1, unfrozen, 10 sl.*	0.672 ± 0.090	0.615 (+0.025)	0.497 (−0.034)
Bbox, λ=0.1, unfrozen, 10 sl.	0.661 ± 0.117	0.539 (−0.051)	0.515 (−0.016)
Bbox, λ=0.0, unfrozen, 10 sl.*	0.657 ± 0.100	0.517 (−0.073)	0.595 (+0.064)
Raw, λ=0.0, frozen, all sl.	0.648 ± 0.100	0.702 (+0.112)	0.515 (−0.016)
Raw, λ=0.1, frozen, all sl.	0.640 ± 0.038	0.635 (+0.045)	0.534 (+0.003)
Bbox, λ=0.0, unfrozen, 10 sl.	0.568 ± 0.101	0.534 (−0.056)	0.494 (−0.037)
Raw, λ=0.1, predefined genes*	0.522 ± 0.087	0.670 (+0.080)	0.477 (−0.054)
Raw, λ=0.1, predefined genes	0.457 ± 0.040	0.574 (−0.016)	0.420 (−0.111)
Radiomics baseline	—	0.590	0.531

Table 3.7 shows the performance of various models on the Soramic and Lausanne cohorts. On the Soramic ablation set, models using a patient-level contrastive split or a frozen backbone with all slices generalise best (AUROC 0.635–0.732) and outperform the radiomic baseline (0.590).

On the Lausanne cohort, overall performance was lower (median AUROC ≈ 0.53), with the majority of models falling near or below the radiomic baseline. Notably, the model that ranked highest on Soramic (frozen backbone, all slices) collapsed to AUROC 0.453 on Lausanne.

These results highlight the challenge of cross-dataset generalisation: CV performance on the resection cohort is a poor predictor of external performance, and architectural choices that work well for one external cohort may not transfer to another. Full classification metrics (AUPRC, sensitivity, specificity, PPV, NPV, and F1) for all models on both cohorts are provided in Appendix B.

Chapter 4

Project Plan

Table 4.1 outlines the planned work items and timeline for the remainder of the project.

Table 4.1: Project plan and timeline.

Timeline	Item	Description
8 Jun – 15 Jun	Late-fusion ensemble for ablation set	Extend the late-fusion ensemble (Section 3.1) to the ablation sets to establish a multimodal upper bound.
8 Jun – 15 Jun	Replace backbone with DINO	Replace the backbone with DINO, run the contrastive learning and evaluate on the Soramic and Lausanne ablation cohorts.
15 Jun – 22 Jul	Mechanistic analysis and distribution analysis	Interpret what the contrastive models have learned and quantify the MRI distribution shift between the resection, Soramic, and Lausanne cohorts.
15 Jun – 5 Jul	Fine-tuning Dino Backbone	Fine-tune the DINO backbone with MRI data, and see if it improves downstream performance.
6 Jul –	Thesis writing	Write up all experiments, analyses, and conclusions in the MSc thesis.

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Appendix A

RNA-seq Feature Selection Details

The tables below list the 20 genes selected by DESeq2 from the full transcriptome in the before-CV run (applied once to the complete task-filtered dataset), along with their Benjamini–Hochberg adjusted p-values. Genes passing BH correction at $p < 0.05$ are marked with an asterisk ($*$); the remaining entries are fallback selections (top 20 by adjusted p-value). For 1-year RFS, six genes pass BH correction; for 2-year RFS, three genes pass. H19 in Table A.2 is the sole gene intersecting the predefined HCC gene set.

Table A.1: DESeq2-selected genes for 1-year RFS prediction (before-CV, full transcriptome, $n = 56$). Asterisk ($*$) denotes BH-adjusted $p < 0.05$.

Gene	BH-adjusted p-value
AL138889.1	5.84×10^{-4} *
AC135731.1	5.84×10^{-4} *
AC004889.1	3.33×10^{-2} *
CCDC26	3.93×10^{-2} *
INAFM2	4.67×10^{-2} *
AL160272.1	4.67×10^{-2} *
AC092068.2	5.43×10^{-2}
LINC00514	5.46×10^{-2}
AL031594.1	9.46×10^{-2}
AL117328.2	1.07×10^{-1}
AC005696.4	1.54×10^{-1}
EGFL8	2.11×10^{-1}
AC025580.2	2.11×10^{-1}
AC006538.1	2.11×10^{-1}
AC127070.4	2.53×10^{-1}
AL353726.2	2.86×10^{-1}
AC016355.1	3.32×10^{-1}
AC008395.1	3.46×10^{-1}
AL008638.3	3.46×10^{-1}
AC102953.2	3.83×10^{-1}

Table A.2: DESeq2-selected genes for 2-year RFS prediction (before-CV, full transcriptome, $n = 54$). Asterisk ($*$) denotes BH-adjusted $p < 0.05$. H19 (†) is the sole intersection with the predefined HCC gene set.

Gene	BH-adjusted p-value
CAMK2N2	8.94×10^{-3} *
AC093826.2	3.18×10^{-2} *
LACC1	4.58×10^{-2} *
CSF2	1.77×10^{-1}
HNRNPA1P9	1.77×10^{-1}
AL445235.1	1.77×10^{-1}
AC025580.2	1.77×10^{-1}
AC025198.1	1.77×10^{-1}
AC093525.8	1.77×10^{-1}
OR52N5	1.97×10^{-1}
RBMXL3	1.97×10^{-1}
AC004241.5	1.97×10^{-1}
AC138647.1	2.23×10^{-1}
AL449283.1	2.73×10^{-1}
AC130366.1	2.73×10^{-1}
AC063947.2	4.55×10^{-1}
H19†	4.63×10^{-1}
SGSM1	4.63×10^{-1}
HIGD2B	4.63×10^{-1}
ZMYND12	4.63×10^{-1}

Appendix B

Complete External Evaluation Metrics

Tables B.1 and B.2 provide full classification metrics for all 16 contrastive models and the radiomic baseline on the Soramic ablation set and the Lausanne cohort for 2-year RFS prediction. Best head (LR or RF) per model is shown, ranked by AUROC. "—" denotes undefined NPV (all samples predicted positive).

Table B.1: Soramic ablation set ($n = 59$, 68% positive): all models ranked by AUROC, 2-year RFS. Best head (LR or RF) per model shown.

Head	Configuration	AUROC	AUPRC	Sens.	Spec.	PPV	NPV	F1
LR	raw, λ=0.1, 2y$_{\text{bCV}}$ genes, n=10, patient	0.732	0.865	0.846	0.389	0.750	0.538	0.795
LR	raw, λ=0.1, frozen, n=all, slice	0.718	0.838	0.846	0.389	0.750	0.538	0.795
LR	raw, λ=0.0, frozen, n=all, patient	0.702	0.840	0.897	0.222	0.714	0.500	0.795
LR	raw, λ=0.0, frozen, n=all, slice	0.684	0.804	0.974	0.278	0.745	0.833	0.844
RF	raw, λ=0.1, predefined genes, n=10, slice	0.670	0.820	0.872	0.278	0.723	0.500	0.791
LR	bbox, λ=0.1, unfrozen, n=10, slice	0.669	0.791	1.000	0.000	0.667	—	0.800
RF	raw, λ=0.1, frozen, n=all, patient	0.635	0.819	1.000	0.000	0.684	—	0.812
LR	raw, λ=0.1, unfrozen, n=10, slice	0.615	0.816	0.385	0.778	0.789	0.368	0.517
RF	raw, λ=0.0, unfrozen, n=10, slice	0.606	0.734	1.000	0.000	0.684	—	0.812
RF	*Radiomic baseline*, 149 arterial features	0.590	0.766	0.143	1.000	1.000	0.375	0.250
RF	bbox, λ=0.1, frozen, n=all, slice	0.577	0.719	0.972	0.000	0.660	0.000	0.787
LR	raw, λ=0.1, predefined genes, n=10, patient	0.574	0.734	0.667	0.500	0.743	0.409	0.703
LR	bbox, λ=0.1, unfrozen, n=10, patient	0.539	0.710	0.944	0.056	0.667	0.333	0.782
LR	bbox, λ=0.0, unfrozen, n=10, patient	0.534	0.661	0.028	0.889	0.333	0.314	0.051
RF	raw, λ=0.1, unfrozen, n=10, patient	0.522	0.707	1.000	0.000	0.684	—	0.812
LR	*Radiomic baseline*, 149 arterial features	0.518	0.671	0.657	0.389	0.676	0.368	0.667
LR	bbox, λ=0.0, unfrozen, n=10, slice	0.517	0.693	0.472	0.556	0.680	0.345	0.557
RF	raw, λ=0.1, 2y$_{\text{bCV}}$ genes, n=10, slice	0.516	0.712	1.000	0.000	0.684	—	0.812
LR	raw, λ=0.0, unfrozen, n=10, patient	0.494	0.674	0.949	0.056	0.685	0.333	0.796

Table B.2: Lausanne cohort ($n = 66$, 74% positive): all models ranked by AUROC, 2-year RFS. Best head (LR or RF) per model shown.

Head	Configuration	AUROC	AUPRC	Sens.	Spec.	PPV	NPV	F1
RF	raw, λ=0.1, unfrozen, n=10, patient	0.771	0.867	1.000	0.000	0.742	—	0.852
RF	bbox, λ=0.1, frozen, n=all, slice	0.614	0.810	0.979	0.059	0.746	0.500	0.847
LR	raw, λ=0.1, $2y_{bCV}$ genes, n=10, slice	0.655	0.850	1.000	0.000	0.742	—	0.852
LR	raw, λ=0.0, unfrozen, n=10, patient	0.618	0.827	0.980	0.059	0.750	0.500	0.850
LR	raw, λ=0.0, unfrozen, n=10, slice	0.600	0.844	0.980	0.059	0.750	0.500	0.850
RF	bbox, λ=0.0, unfrozen, n=10, slice	0.595	0.790	0.438	0.706	0.808	0.308	0.568
LR	raw, λ=0.1, $2y_{bCV}$ genes, n=10, patient	0.563	0.806	0.694	0.353	0.756	0.286	0.723
RF	raw, λ=0.0, frozen, n=all, slice	0.556	0.803	0.959	0.059	0.746	0.333	0.839
RF	bbox, λ=0.1, unfrozen, n=10, slice	0.544	0.755	0.333	0.765	0.800	0.289	0.471
LR	*Radiomic baseline*, 149 arterial features	0.531	0.748	0.422	0.625	0.760	0.278	0.543
LR	raw, λ=0.1, frozen, n=all, patient	0.534	0.775	0.939	0.118	0.754	0.400	0.836
RF	bbox, λ=0.1, unfrozen, n=10, patient	0.515	0.769	1.000	0.000	0.738	—	0.850
LR	raw, λ=0.0, frozen, n=all, patient	0.515	0.779	0.755	0.294	0.755	0.294	0.755
RF	*Radiomic baseline*, 149 arterial features	0.506	0.747	0.067	0.875	0.600	0.250	0.120
LR	raw, λ=0.1, unfrozen, n=10, slice	0.497	0.753	0.204	0.765	0.714	0.250	0.317
RF	bbox, λ=0.0, unfrozen, n=10, patient	0.494	0.766	0.646	0.353	0.738	0.261	0.689
RF	raw, λ=0.1, frozen, n=all, slice	0.453	0.727	0.776	0.118	0.717	0.154	0.745
RF	raw, λ=0.1, predefined genes, n=10, slice	0.477	0.755	0.796	0.176	0.736	0.231	0.765
LR	raw, λ=0.1, predefined genes, n=10, patient	0.420	0.702	0.531	0.412	0.722	0.233	0.612

Appendix C

RNA-seq Baseline Full Results

Three configurations were evaluated, all using DESeq2 feature selection (BH-adjusted $p < 0.05$; fallback to top 20 genes by adjusted p-value), 3-fold stratified CV, L_1 logistic regression ($C \in \{0.001, 0.01, 0.1, 1\}$) and random forest (depth $\in \{2, 4\}$, min leaf $\in \{5, 10, 15\}$):

C1 `before_cv`, full transcriptome (50,986 genes) — DESeq2 fit once on the full labelled dataset before CV (information leak; diagnostic only).

C2 `before_cv`, predefined HCC gene set (2,146 genes) — same leakage setting but restricted to the curated HCC prior.

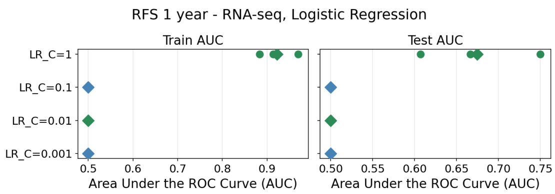
C3 `in_cv`, predefined HCC gene set — statistically valid: DESeq2 refit inside each training fold.

Table C.1 summarises the best mean test AUROC per classifier.

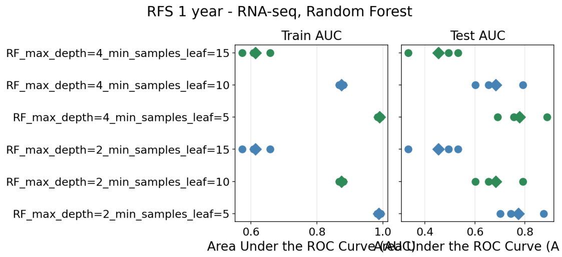
Table C.1: Best mean test AUROC per configuration and classifier for RNA-seq baselines. C1 values are leakage-inflated (before-CV selection). C2 and C3 collapse to majority-class prediction because no gene survives BH correction.

Config	1y LR	1y RF	2y LR	2y RF
C1 — before-CV, all genes	**0.686**	0.601	**0.864**	0.761
C2 — before-CV, predefined	0.500	0.570	0.500	0.505
C3 — in-CV, predefined	0.500	0.500	0.500	0.508

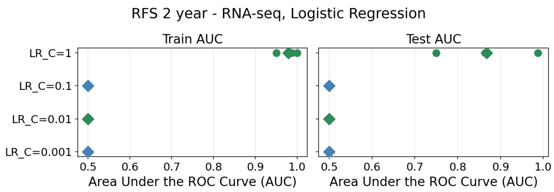
C1 — Before-CV, Full Transcriptome



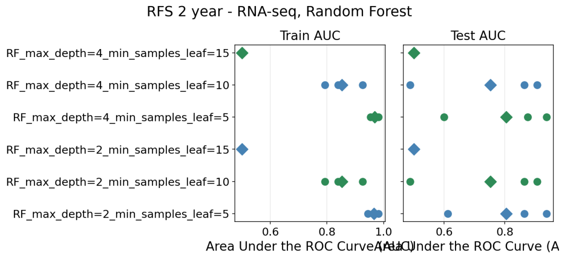
(a) 1-year RFS, LR



(b) 1-year RFS, RF



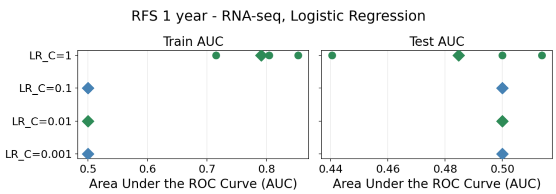
(c) 2-year RFS, LR



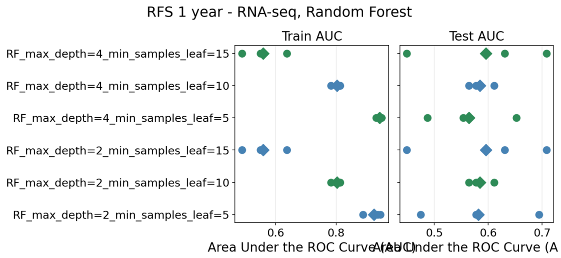
(d) 2-year RFS, RF

Figure C.1: C1 (before-CV, full transcriptome): 3-fold CV AUROC strip plots. Circles = per-fold AUC; diamonds = mean. Values are leakage-inflated.

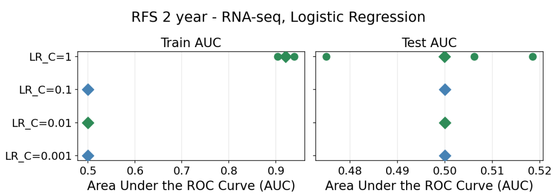
C2 — Before-CV, Predefined HCC Genes



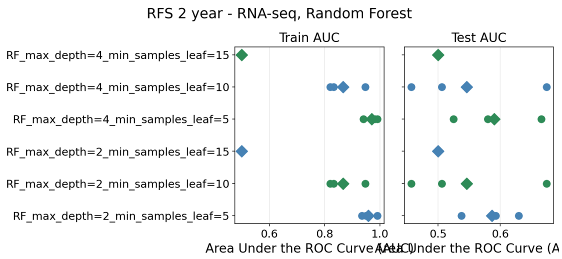
(a) 1-year RFS, LR



(b) 1-year RFS, RF



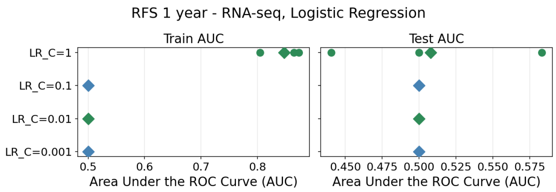
(c) 2-year RFS, LR



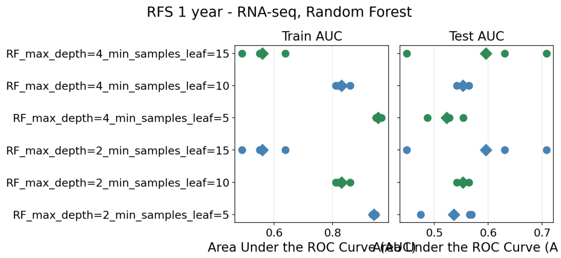
(d) 2-year RFS, RF

Figure C.2: C2 (before-CV, predefined HCC genes): all models collapse to near-chance AUROC because no gene survives BH correction; every fold uses the top-20 fallback.

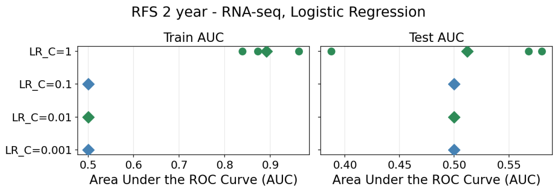
C3 — In-CV, Predefined HCC Genes



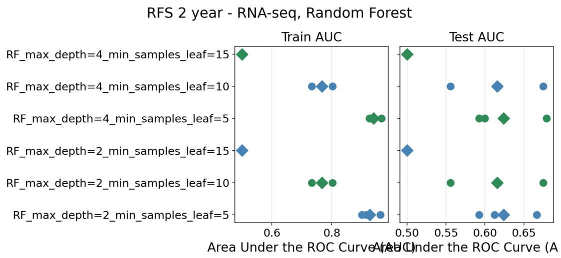
(a) 1-year RFS, LR



(b) 1-year RFS, RF



(c) 2-year RFS, LR



(d) 2-year RFS, RF

Figure C.3: C3 (in-CV, predefined HCC genes): the statistically valid configuration. Performance matches C2, confirming the predefined gene set carries no detectable differential expression signal at this cohort size.